1. Solve:

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0,$$

$$u\Big|_{x=0}=u_0, \qquad u\Big|_{x=a}=u_0\left[3\left(\frac{y}{b}\right)^2-2\left(\frac{y}{b}\right)^3\right],$$

$$\frac{\partial u}{\partial y}\Big|_{y=0}=0, \qquad \frac{\partial u}{\partial y}\Big|_{y=b}=0,$$

( b=constant)

$u(x,y)=X(x)Y(y) \Rightarrow X''(x)Y(y)+X(x)Y''(y)=0$，$\begin{cases} X''(x)-\lambda X(x)=0 \\ Y''(y)+\lambda Y(y)=0 \end{cases}$ $\frac{du}{dy}\Big|_{y=0}=0,\ \frac{du}{dy}\Big|_{y=b}=0$

当 $\lambda=0$ $Y''(y)=0 \Rightarrow Y(y)=Ay+B$ 代回可知 $A=0$ 不正确

当 $\lambda<0$ $\lambda=-s^2$ $s=\sqrt{-\lambda}$. $Y(y)=Ae^{sy}+Be^{-sy}$ $Y'(y)=Ase^{sy}-Bse^{-sy}$ $Y'(0)=As-Bs=0$ $A=B$ $Y'(b)=As(e^{sb}-e^{-sb})=0$ A:
不正确

当 $\lambda>0$ $\lambda=\alpha^2$ $Y(y)=A\cos\sqrt{\lambda}y+B\sin\sqrt{\lambda}y$ 则有 $Y'(0)=B\sqrt{\lambda}=Y'(b)=-A\sqrt{\lambda}\sin\sqrt{\lambda}b+B\sqrt{\lambda}\cos\sqrt{\lambda}b=0 \Rightarrow B=0$ 则 $A\sqrt{\lambda}\sin\sqrt{\lambda}b=0$

为免 $Y(y)=0$ 则 $A\neq 0$, $\sin\sqrt{\lambda}b=0$. 可知 $\lambda=(\frac{n\pi}{b})^2$. 且 $Y_n(y)=A_n\cos(\frac{n\pi y}{b})$. (取 $A=1$)

则代回有 $X''(x)-(\frac{n\pi}{b})^2X(x)=0$ $X_n(x)=C_n\sinh(\frac{n\pi x}{b})+D_n\cosh(\frac{n\pi x}{b})$

则有一般解 $u(x,y)=\sum\limits_{n=1}^{\infty}[C_n\sinh(\frac{n\pi x}{b})+D_n\cosh(\frac{n\pi x}{b})]\cos(\frac{n\pi y}{b})$ 代入 $\begin{cases} X(0)Y(y)=u_0 \\ X(a)Y(y)=u_0[3(\frac{y}{b})^2-2(\frac{y}{b})^3] \end{cases}$

$\Rightarrow \begin{cases} \sum\limits_{n=1}^{\infty}\cos\frac{n\pi}{b}y\ D_n=u_0 \\ \sum\limits_{n=1}^{\infty}(C_n\sinh\frac{n\pi a}{b}+D_n\cosh\frac{n\pi a}{b})\cos\frac{n\pi}{b}y=u_0[3(\frac{y}{b})^2-2(\frac{y}{b})^3] \end{cases}$ 利用本征函数正交性, $D_n=\frac{2}{b}\int_0^b u_0\cos\frac{n\pi}{b}y\,dy$.

$\begin{cases} X_n''(x)-\lambda_n X_n(x)=0 \\ X_n(0)=a_n,\ X_n(a)=b_n \end{cases} \Rightarrow a_n=\dfrac{\int_0^b u_0 Y_n(y)dy}{\int_0^b Y_n^2(y)dy}=0$ $b_n=\dfrac{\int_0^b u_0[3(\frac{y}{b})^2-2(\frac{y}{b})^3]Y_n(y)dy}{\int_0^b Y_n^2(y)dy}=\begin{cases} 0,\ n 为偶 \\ -\frac{48u_0}{\lambda_n^2 b^4},\ n 为奇 \end{cases}$

则 $X_n=\begin{cases} 0 &, n 为偶 \\ -\dfrac{48u_0}{(n\pi)^4\sinh\frac{n\pi}{b}a}\sin\frac{n\pi}{b}x &, n 为奇 \end{cases}$ $\Rightarrow u(x,y)=u_0(1-\frac{x}{2a})-\frac{48u_0}{\pi^4}\sum\limits_{n=1}^{\infty}\frac{1}{(2n+1)^4}\frac{\sinh\frac{2n+1}{b}\pi x}{\sinh\frac{2n+1}{b}\pi a}\cos\frac{2n+1}{b}\pi y.$

2. Solve:

$$\frac{\partial^2 u}{\partial t^2}-a^2\frac{\partial^2 u}{\partial x^2}=bx(l-x),$$

$$u\Big|_{x=0}=0, \qquad u\Big|_{x=l}=0,$$

$$u\Big|_{t=0}=0, \qquad \frac{\partial u}{\partial t}\Big|_{t=0}=0.$$

设方程一个特解为 $v(x)$，则 $v''=-\dfrac{b}{a^2}x(l-x)$，使之满足齐次边界条件，解之得

$v=\dfrac{b}{12a^2}x\left(x^3-2lx^2+l^3\right)$。设 $u=v+w$，则 $w$ 满足

$$\begin{cases} \dfrac{\partial^2 w}{\partial t^2}-a^2\dfrac{\partial^2 w}{\partial x^2}=0 \\ w\Big|_{x=0}=0, w\Big|_{x=l}=0 \\ w\Big|_{t=0}=-\dfrac{b}{12a^2}x\left(x^3-2lx^2+l^3\right), \dfrac{\partial w}{\partial t}\Big|_{t=0}=0 \end{cases}$$ 。

$$w(x,t)=\sum_{n=1}^{\infty}\left(A_n\sin\frac{n\pi}{l}at+B_n\cos\frac{n\pi}{l}at\right)\sin\frac{n\pi}{l}x,$$

$A_n=0$，$B_n=-\dfrac{b}{6a^2l}\int_0^l x\left(x^3-2lx^2+l^3\right)\sin\dfrac{n\pi}{l}xdx=\dfrac{4l^4b}{n^5\pi^5a^2}\left[(-1)^n-1\right]$，

所以 $u(x,t)=\dfrac{b}{12a^2}x\left(x^3-2lx^2+l^3\right)$

$$-\frac{8l^4b}{\pi^5a^2}\sum_{n=1}^{\infty}\frac{1}{(2k+1)^5}\cos\frac{(2k+1)\pi}{l}at\sin\frac{(2k+1)\pi}{l}x。$$

3. Solve in rectanglar region : $0 \leqslant x \leqslant a,\ -b/2 \leqslant y \leqslant b/2$

$$(1)\begin{cases}\nabla^2 u=-2,\\ u\big|_{x=0,a}=0,\\ u\big|_{y=\pm b/2}=0;\end{cases}\qquad (2)\begin{cases}\nabla^2 u=-x^2y,\\ u\big|_{x=0,a}=0,\\ u\big|_{y=\pm b/2}=0.\end{cases}$$

$u\big|_{y=\pm b/2}=0$ $u\big|_{y=\pm b/2}=0.$

(1) 则 $\begin{cases}\frac{d^2u}{dx^2}+\frac{d^2u}{dy^2}=-2\\ u|_{x=0}=u|_{x=a}=0\\ u|_{y=\frac{b}{2}}=u|_{y=-\frac{b}{2}}=0\end{cases}$ 令 $u(x,y)=v(x)+w(x,y)$ 其中 $v(x)$ 为满足 $\frac{d^2v}{dx^2}+\frac{d^2v}{dy^2}=-2$. $v|_{x=0,a}=0$ 的特解

可知 $v(x)=-x^2+ax$

又有 $\begin{cases}\frac{d^2w}{dx^2}+\frac{d^2w}{dy^2}=0\\ w|_{x=0,a}=0\\ w|_{y=\pm\frac{b}{2}}=-v(x)\end{cases}$ 求得一般解 $u(x,t)=v(x)+\sum_{n=1}^{\infty}(C_ne^{\frac{n\pi y}{a}}+D_ne^{\frac{-n\pi y}{a}})\sin\frac{n\pi}{a}x$，又代入 $w|_{y=\pm\frac{b}{2}}=-v(x)$ 可得

将 $v(x)$ 展开为傅里叶级数 $x(x-a)=\sum_{n=1}^{m}A_n\sin\frac{n\pi x}{a}$ $A_n=\frac{2}{a}\int_0^a(x^2-ax)\sin\frac{n\pi x}{a}dx=\frac{4a^2}{n^3\pi^3}[(-1)^n-1]$

$$\begin{cases}C_ne^{-\frac{n\pi b}{2a}}+D_ne^{\frac{n\pi b}{2a}}=A_n\\ C_ne^{\frac{n\pi b}{2a}}+D_ne^{-\frac{n\pi b}{2a}}=A_n\end{cases}\Rightarrow D_n=\frac{\sinh(\frac{n\pi b}{2a})}{\sinh(\frac{n\pi b}{a})}A_n.\quad C_n=\frac{\sinh(\frac{n\pi b}{2a})}{\sinh(\frac{n\pi b}{a})}A_n.$$

可知 $u(x,y)=-x^2+ax-\frac{8a^2}{\pi^3}\sum_{n=0}^{\infty}\frac{1}{(2n+1)^3}\frac{\cosh\frac{2n+1}{a}\pi y}{\cosh\frac{2n+1}{2a}\pi b}\sin\frac{2n+1}{a}\pi x$

(2) 同理 令 $u(x,y)=w(x,y)+v(x,y)$. 可求得 $v(x,y)=-\frac{1}{12}xy(x^3-a^3)$

对 $w(x,y)$ 有 $\begin{cases}\frac{d^2w}{dx^2}+\frac{d^2w}{dy^2}=0\\ w|_{x=0}=0\quad w|_{x=a}=0\\ w|_{y=\pm\frac{b}{2}}=-v(x,y)\end{cases}\Rightarrow u(x,y)=\sum_{n=1}^{\infty}(A_ne^{\frac{n\pi y}{a}}+B_ne^{\frac{-n\pi y}{a}})\sin\frac{n\pi}{a}x$

$u(x,\frac{b}{2})=\sum_{n=1}^{\infty}(A_ne^{\frac{n\pi b}{2a}}+B_ne^{-\frac{n\pi b}{2a}})\sin\frac{n\pi}{a}x=\frac{bx}{24}(x^3-a^3)$ $u(x,-\frac{b}{2})=\sum_{n=1}^{\infty}(A_ne^{-\frac{n\pi b}{2a}}+B_ne^{\frac{n\pi b}{2a}})\sin\frac{n\pi}{a}x=-\frac{bx}{24}(x^3-a^3)$

傅里叶展开 $\frac{bx}{24}(x^3-a^3)=\sum_{n=1}^{m}C_n\sin\frac{n\pi}{a}x$ $C_n=\frac{2}{a}\int_0^a\frac{bx}{24}(x^3-a^3)\sin\frac{n\pi x}{a}dx=\frac{2a^4b}{n^3\pi^3}(\frac{1}{2}(-1)^n)-\frac{2a^4b}{n^5\pi^5}[(-1)^n-1]$

$$\begin{cases}A_ne^{\frac{n\pi b}{2a}}+B_ne^{-\frac{n\pi b}{2a}}=C_n\\ A_ne^{-\frac{n\pi b}{2a}}+B_ne^{\frac{n\pi b}{2a}}=-C_n\end{cases}\Rightarrow \text{解得 } A_n=B_n=\frac{\cosh(\frac{n\pi b}{2a})}{\sinh(\frac{n\pi b}{a})}C_n$$

则可知 $u(x,y)=-\frac{1}{12}xy(x^3-a^3)+\frac{a^4b}{\pi^3}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^3}\frac{\sinh\frac{n\pi}{a}y}{\sinh\frac{n\pi}{2a}y}\sin\frac{n\pi}{a}x+\frac{4a^4b}{\pi^5}\sum_{n=0}^{\infty}(\frac{1}{2n+1})^5\frac{\sinh(\frac{2n+1}{a}\pi y)}{\sinh(\frac{2n+1}{2a}\pi b)}\sin\frac{2n+1}{a}\pi x$

另外的解法:

证明. Let $u(x,y)=X(x)Y(y)$. Then the original equation becomes

$$\begin{cases}X''(x)Y(y)+X(x)Y''(y)=-2\\ X(0)Y(y)=X(a)Y(y)=0\\ X(x)Y(\frac{b}{2})=X(x)Y(-\frac{b}{2})=0.\end{cases}$$

So we can consider the eigenvalue problem

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(a) = 0, \end{cases}$$

which has the solution $X_n(x) = \sin\sqrt{\lambda_n}x$ with $\lambda_n = \left(\frac{n\pi}{a}\right)^2$ ($n \in \mathbb{N}$). Suppose $u(x,y) = \sum_{n=1}^{\infty} X_n(x)Y_n(y)$. Then by the orthogonality of $\{X_n(x)\}_{n=1}^{\infty}$, we have

$$\begin{cases} Y_n''(y) - \lambda_n Y_n(y) = \frac{4}{n\pi}[(-1)^n - 1] \\ Y_n(-\frac{b}{2}) = Y_n(\frac{b}{2}) = 0. \end{cases}$$

To find the expression for $Y_n(y)$, we note the Green's function associated with $Y_n(y)$ satisfies

$$\begin{cases} \frac{d^2}{dy^2}g(y;t) - \lambda_n g(y;t) = \delta(y-t) \\ g(-\frac{b}{2};t) = g(\frac{b}{2};t) = 0. \end{cases}$$

Therefore

$$g(y;t) = \begin{cases} A(t)\sinh\sqrt{\lambda_n}(y+\frac{b}{2}), & -\frac{b}{2} < y < t \\ B(t)\sinh\sqrt{\lambda_n}(y-\frac{b}{2}), & t < y < \frac{b}{2}, \end{cases}$$

where the coefficient function $A(t)$ and $B(t)$ are determined by the continuity property of $g(y;t)$ at $y = t$:

$$\begin{cases} A(t)\sinh\sqrt{\lambda_n}(t+\frac{b}{2}) = B(t)\sinh\sqrt{\lambda_n}(t-\frac{b}{2}) \\ \sqrt{\lambda_n}B(t)\cosh\sqrt{\lambda_n}(t-\frac{b}{2}) - \sqrt{\lambda_n}A(t)\cosh\sqrt{\lambda_n}(t+\frac{b}{2}) = 1. \end{cases}$$

Solving this equation gives

$$\begin{cases} A(t) = \frac{1}{\sqrt{\lambda_n}\sinh\sqrt{\lambda_n}b}\sinh\sqrt{\lambda_n}(t-\frac{b}{2}) \\ B(t) = \frac{1}{\sqrt{\lambda_n}\sinh\sqrt{\lambda_n}b}\sinh\sqrt{\lambda_n}(t+\frac{b}{2}). \end{cases}$$

Therefore

$$g(y;t) = \frac{\sinh\sqrt{\lambda_n}(t-\frac{b}{2})\sinh\sqrt{\lambda_n}(y+\frac{b}{2})1_{\{-\frac{b}{2}<y<t\}} + \sinh\sqrt{\lambda_n}(t+\frac{b}{2})\sinh\sqrt{\lambda_n}(y-\frac{b}{2})1_{\{t<y<\frac{b}{2}\}}}{\sqrt{\lambda_n}\sinh\sqrt{\lambda_n}b}$$

and

$$\begin{aligned} \int_{-\infty}^{\infty} g(y;t)dt &= \int_{-\frac{b}{2}}^{y} \frac{\sinh\sqrt{\lambda_n}(y-\frac{b}{2})}{\sqrt{\lambda_n}\sinh\sqrt{\lambda_n}b}\sinh\sqrt{\lambda_n}(t+\frac{b}{2})dt + \int_{y}^{\frac{b}{2}} \frac{\sinh\sqrt{\lambda_n}(y+\frac{b}{2})}{\sqrt{\lambda_n}\sinh\sqrt{\lambda_n}b}\sinh\sqrt{\lambda_n}(t-\frac{b}{2})dt \\ &= \frac{\sinh\sqrt{\lambda_n}(y-\frac{b}{2})}{\lambda_n\sinh\sqrt{\lambda_n}b}\left[\cosh\sqrt{\lambda_n}(y+\frac{b}{2}) - 1\right] + \frac{\sinh\sqrt{\lambda_n}(y+\frac{b}{2})}{\lambda_n\sinh\sqrt{\lambda_n}b}\left[1 - \cosh\sqrt{\lambda_n}(y-\frac{b}{2})\right] \\ &= \frac{-\sinh\sqrt{\lambda_n}b - \sinh\sqrt{\lambda_n}(y-\frac{b}{2}) + \sinh\sqrt{\lambda_n}(y+\frac{b}{2})}{\lambda_n\sinh\sqrt{\lambda_n}b} \\ &= \frac{-\sinh\sqrt{\lambda_n}b + 2\sinh\frac{\sqrt{\lambda_n}b}{2}\cosh\sqrt{\lambda_n}y}{\lambda_n\sinh\sqrt{\lambda_n}b} \\ &= -\frac{1}{\lambda_n} + \frac{\cosh\sqrt{\lambda_n}y}{\lambda_n\cosh\frac{\sqrt{\lambda_n}b}{2}}. \end{aligned}$$

Hence

$$\begin{aligned} Y_n(y) &= \frac{4}{n\pi}[(-1)^n-1]\int_{-\infty}^{\infty} g(y;t)dt = \frac{4}{n\pi}[(-1)^{n+1}+1]\left[\frac{a^2}{(n\pi)^2} - \frac{a^2\cosh\frac{n\pi y}{a}}{(n\pi)^2\cosh\frac{n\pi b}{2a}}\right] \\ &= \begin{cases} \frac{8a^2}{(n\pi)^3}\left[1-\frac{\cosh\frac{n\pi y}{a}}{\cosh\frac{n\pi b}{2a}}\right] & \text{if } n \text{ is odd,} \\ 0 & \text{if } n \text{ is even.} \end{cases} \end{aligned}$$

Combined, we have

$$u(x,t) = \sum_{n=1}^{\infty} X_n(x)Y_n(y) = \frac{8a^2}{\pi^3}\sum_{n=0}^{\infty}\frac{1}{(2n+1)^3}\left[1-\frac{\cosh\frac{2n+1}{a}\pi y}{\cosh\frac{2n+1}{2a}\pi b}\right]\sin\frac{2n+1}{a}\pi x.$$

(2)

证明. Similar to part (1), we find the solution to the eigenvalue problem

$$\begin{cases} X''(x)+\lambda X(x)=0 \\ X(0)=X(a)=0 \end{cases}$$

as $X_n(x)=\sin\sqrt{\lambda_n}x$ with $\lambda_n=\left(\frac{n\pi}{a}\right)^2$ $(n\in\mathbb{N})$. Expanding $u(x,y)$ according to $\{X_n(x)\}_{n=1}^{\infty}$, we have

$$u(x,y) = \sum_{n=1}^{\infty} Y_n(y)X_n(x),$$

where

$$Y_n(y) = \frac{\int_0^a x^2yX_n(x)dx}{\int_0^a X_n^2(x)dx} = \frac{2}{a}\int_0^a x^2yX_n(x)dx = 2y\left\{\frac{a^2(-1)^{n+1}}{n\pi}+\frac{2a^2}{(n\pi)^3}[(-1)^n-1]\right\}.$$

So the eigenvalue problem associated with $Y_n(x)$ becomes

$$\begin{cases} Y_n''(x)-\lambda_nY_n(x)=2y\left\{\frac{a^2(-1)^{n+1}}{n\pi}+\frac{2a^2}{(n\pi)^3}[(-1)^n-1]\right\} \\ Y_n(-\frac{b}{2})=Y_n(\frac{b}{2})=0. \end{cases}$$

The Green's function associated with $Y_n(y)$ is the same as that of part (1):

$$g(y;t) = \frac{\sinh\sqrt{\lambda_n}(t-\frac{b}{2})\sinh\sqrt{\lambda_n}(y+\frac{b}{2})1_{\{-\frac{b}{2}<y<t\}}+\sinh\sqrt{\lambda_n}(t+\frac{b}{2})\sinh\sqrt{\lambda_n}(y-\frac{b}{2})1_{\{t<y<\frac{b}{2}\}}}{\sqrt{\lambda_n}\sinh\sqrt{\lambda_n}b}.$$

Therefore

$$\begin{aligned} Y_n(y) &= \int_{-\infty}^{\infty} 2t\left\{\frac{a^2(-1)^{n+1}}{n\pi}+\frac{2a^2}{(n\pi)^3}[(-1)^n-1]\right\}g(y;t)dt \\ &= \frac{2a^2}{n\pi}\left\{(-1)^{n+1}+\frac{2}{(n\pi)^2}[(-1)^n-1]\right\}\int_{-\infty}^{\infty} tg(y;t)dt. \end{aligned}$$

Note

$$
\begin{aligned}
\int_{-\infty}^{\infty} tg(y;t)dt &= \frac{\sinh\sqrt{\lambda_n}(y+\frac{b}{2})}{\sqrt{\lambda_n}\sinh\sqrt{\lambda_n}b}\int_y^{\frac{b}{2}} t\sinh\sqrt{\lambda_n}(t-\frac{b}{2})dt + \frac{\sinh\sqrt{\lambda_n}(y-\frac{b}{2})}{\sqrt{\lambda_n}\sinh\sqrt{\lambda_n}b}\int_{-\frac{b}{2}}^{y} t\sinh\sqrt{\lambda_n}(t+\frac{b}{2})dt \\
&= \frac{\sinh\sqrt{\lambda_n}(y+\frac{b}{2})}{\sqrt{\lambda_n}\sinh\sqrt{\lambda_n}b}\left[\frac{b}{2\sqrt{\lambda_n}} - \frac{y}{\sqrt{\lambda_n}}\cosh\sqrt{\lambda_n}(y-\frac{b}{2}) + \frac{1}{\lambda_n}\sinh\sqrt{\lambda_n}(y-\frac{b}{2})\right] + \\
&\quad \frac{\sinh\sqrt{\lambda_n}(y-\frac{b}{2})}{\sqrt{\lambda_n}\sinh\sqrt{\lambda_n}b}\left[\frac{b}{2\sqrt{\lambda_n}} + \frac{y\cosh\sqrt{\lambda_n}(y+\frac{b}{2})}{\sqrt{\lambda_n}} - \frac{1}{\lambda_n}\sinh\sqrt{\lambda_n}(y+\frac{b}{2})\right] \\
&= \frac{b[\sinh\sqrt{\lambda_n}(y+\frac{b}{2}) + \sinh\sqrt{\lambda_n}(y-\frac{b}{2})]}{2\lambda_n\sinh\sqrt{\lambda_n}b} + \frac{y\sinh(-b)\sqrt{\lambda_n}}{\lambda_n\sinh\sqrt{\lambda_n}b} \\
&= \frac{b}{2\lambda_n}\frac{\sinh\sqrt{\lambda_n}y}{\sinh\frac{\sqrt{\lambda_n}b}{2}} - \frac{y}{\lambda_n}.
\end{aligned}
$$

Therefore

$$
\begin{aligned}
Y_n(y) &= \frac{2a^2}{n\pi}\left\{(-1)^{n+1} + \frac{2}{(n\pi)^2}[(-1)^n - 1]\right\}\left(\frac{b}{2\lambda_n}\frac{\sinh\sqrt{\lambda_n}y}{\sinh\frac{\sqrt{\lambda_n}b}{2}} - \frac{y}{\lambda_n}\right) \\
&= \begin{cases} -\frac{2a^2}{n\pi\lambda_n}\left(y - \frac{b}{2}\frac{\sinh\sqrt{\lambda_n}y}{\sinh\frac{\sqrt{\lambda_n}b}{2}}\right) + \frac{4}{(n\pi)^2}\frac{2a^2}{n\pi\lambda_n}\left(y - \frac{b}{2}\frac{\sinh\sqrt{\lambda_n}y}{\sinh\frac{\sqrt{\lambda_n}b}{2}}\right) & \text{if } n \text{ is odd,} \\ \frac{2a^2}{n\pi\lambda_n}\left(y - \frac{b}{2}\frac{\sinh\sqrt{\lambda_n}y}{\sinh\frac{\sqrt{\lambda_n}b}{2}}\right) & \text{if } n \text{ is even.} \end{cases}
\end{aligned}
$$

Hence

$$
\begin{aligned}
&u(x,y) \\
=& \sum_{n=1}^{\infty} Y_n(y)X_n(x) \\
=& \frac{2a^4}{\pi^3}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^3}\left(y - \frac{b}{2}\frac{\sinh\frac{n\pi}{a}y}{\sinh\frac{n\pi}{2a}b}\right)\sin\frac{n\pi}{a}x + \frac{8a^4}{\pi^5}\sum_{n=0}^{\infty}\frac{1}{(2n+1)^5}\left(y - \frac{b}{2}\frac{\sin\frac{2n+1}{a}\pi y}{\sinh\frac{2n+1}{2a}\pi b}\right)\sin\frac{2n+1}{a}\pi x.
\end{aligned}
$$

**注 26.** *The above solution differs from the textbook's answer by a sign. Check.*

4. A slender beam is fixed at x=0, while the end at x=l is subjected to a periodic force Asin ωt. Assuming the initial displacement and initial velocity are both zero, solve the transverse vibration problem of the beam.

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} = a^2 \dfrac{\partial^2 u}{\partial x^2} & t>0,\ 0<x<l \\ u|_{x=0}=0,\ u|_{x=l} = A\sin \omega t & \omega \neq \dfrac{an\pi}{l} \\ u|_{t=0}=0,\ \dfrac{\partial u}{\partial t}\Big|_{t=0}=0 & \end{cases}$$

$$u(x,t) = v(x,t) + w(x,t)$$

$$v(x,t) = A(x)\sin \omega t$$

$$A''(x) + \frac{\omega^2}{a^2} A(x) = 0$$

$$A(0)\sin \omega t = 0, \quad A(l)\sin \omega t = A\sin \omega t$$

$$A(0) = 0, \quad A(l) = A$$

$$A(x) = A\frac{\sin\frac{\omega x}{a}}{\sin\frac{\omega l}{a}} \qquad v(x,t) = A\frac{\sin\frac{\omega x}{a}}{\sin\frac{\omega l}{a}}\sin \omega t$$

$$\begin{cases} \dfrac{\partial^2 w}{\partial t^2} - a^2\dfrac{\partial^2 w}{\partial x^2} = 0 \\ w|_{x=0}=0, \quad w|_{x=l}=0 \\ w|_{t=0}=0, \quad \dfrac{\partial w}{\partial t}\Big|_{t=0} = -A\omega\dfrac{\sin\frac{\omega x}{a}}{\sin\frac{\omega l}{a}} \end{cases}$$

$$w(t,x) = 2A\omega a l \sum_{n=1}^{+\infty} \frac{(-1)^{n+1}}{(\omega l)^2 - (n\pi a)^2}\sin\frac{n\pi a t}{l}\sin\frac{n\pi x}{l}$$

$$u(x,t) = v(x,t) + w(x,t)$$

5. Try to solve the following solution problem

$$\frac{\partial^2 u}{\partial t^2} - a^2\frac{\partial^2 u}{\partial x^2} = 0,$$
$$u\big|_{x=0} = \cos\frac{\pi}{l}at, \qquad \frac{\partial u}{\partial x}\Big|_{x=l} = 0,$$
$$u\big|_{t=0} = \cos\frac{\pi}{l}x, \qquad \frac{\partial u}{\partial t}\Big|_{t=0} = \sin\frac{\pi}{2l}x.$$

*Proof.* We choose $v(x,t) = \cos\frac{\pi}{l}x\cos\frac{\pi}{l}at$ and suppose $u(x,t) = v(x,t) + w(x,t)$. Then $w(x,t)$ satisfies the equation

$$\begin{cases} \frac{\partial^2 w}{\partial t^2} - a^2\frac{\partial^2 w}{\partial x^2} = 0 & \\ w|_{x=0} = 0, & \frac{\partial w}{\partial x}\big|_{x=l} = 0 \\ w(x,0) = 0, & \frac{\partial w}{\partial t}\big|_{t=0} = \sin\frac{\pi}{2l}x. \end{cases}$$

Solving the associated eigenvalue problem

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = 0, X'(l) = 0 \end{cases}$$

gives the eigenfunctions $\{X_n(x)\} = \{\sin\sqrt{\lambda_n}x\}$, with $\lambda_n = \left(\frac{\pi+2n\pi}{2l}\right)^2$. Suppose $w(x,t) = \sum_{n=0}^{\infty} T_n(t)X_n(x)$, then we must have

$$\begin{cases} \sum_{n=0}^{\infty}(T_n''(t) + \lambda_n a^2)X_n(x) = 0 \\ \sum_{n=0}^{\infty} T_n(0)X_n(x) = 0,\ \sum_{n=0}^{\infty} T_n'(0)X_n(x) = X_0(x). \end{cases}$$

Therefore $T_n(t) \equiv 0$ for $n \geq 1$ and $T_0(t) = \frac{2l}{a\pi}\sin\frac{\pi}{2l}at$. Combined, we conclude $u(x,t) = \cos\frac{\pi}{l}x\cos\frac{\pi}{l}at + \frac{2l}{\pi a}\sin\frac{\pi}{2l}x\sin\frac{\pi}{2l}at$. □

6. Try to solve the following solution problem

$$\frac{\partial u}{\partial t} - \kappa\frac{\partial^2 u}{\partial x^2} = 0,$$

$$u\big|_{x=0} = Ae^{i\omega t}, \qquad u\big|_{x=l} = 0,$$

$$u\big|_{t=0} = 0.$$

解：▲令 $u(x,t) = V(x,t) + W(x,t)$. $V(x,t)$为特解，设为$V(x,t) = A(x)e^{j\omega t}$.

代入方程得 $j\omega A(x)e^{j\omega t} - ke^{j\omega t}A''(x) = 0$.

$$A''(x) - \frac{j\omega}{k}A(x) = 0. \quad A(0) = A,\ A(l) = 0.$$

设$A(x) = k_1\sinh\left(\sqrt{\frac{j\omega}{k}}x\right) + k_2\cosh\left(\sqrt{\frac{j\omega}{k}}x\right)$, $A(0) = k_2 = A$.

$$A(l) = k_1\sinh\left(\sqrt{\frac{j\omega}{k}}l\right) + A\cosh\left(\sqrt{\frac{j\omega}{k}}l\right) = 0 \qquad k_1 = \frac{-A\cosh\left(\sqrt{\frac{j\omega}{k}}l\right)}{\sinh\left(\sqrt{\frac{j\omega}{k}}l\right)}$$

$$\therefore \boxed{V(x,t) = \left[\frac{-A\cosh\left(\sqrt{\frac{j\omega}{k}}l\right)}{\sinh\left(\sqrt{\frac{j\omega}{k}}l\right)}\sinh\left(\sqrt{\frac{j\omega}{k}}x\right) + A\cosh\left(\sqrt{\frac{j\omega}{k}}x\right)\right]e^{j\omega t}.}$$

▲ $W(x,t) = X(x)\cdot T(t)$. $W|_{x=0} = 0$, $W|_{x=l} = 0$, $W|_{t=0} = -V(x,0)$

$$X(x)T'(t) - kX''(x)T(t) = 0 \qquad \frac{X''(x)}{X(x)} = \frac{1}{k}\frac{T'(x)}{T(x)} = -\lambda. \qquad T'(x) + \lambda kT(x) = 0.$$

• 本征值问题：$X''(x) + \lambda X(x) = 0$. $X(0) = 0$, $X(l) = 0$.

$$\lambda = \left(\frac{n\pi}{l}\right)^2, \quad X(x) = \sin\left(\frac{n\pi}{l}x\right), \quad n = 1, 2, \cdots$$

• $T_n(t) = C_ne^{-\lambda kt} = C_ne^{-\left(\frac{n\pi}{l}\right)^2kt}$ $\qquad W_n(x,t) = \sin\left(\frac{n\pi}{l}x\right)\cdot C_n\cdot e^{-\left(\frac{n\pi}{l}\right)^2kt}$

$$W|_{t=0} = C_n\sin\left(\frac{n\pi}{l}x\right) = -k_1\sinh\left(\sqrt{\frac{j\omega}{k}}x\right) - k_2\cosh\left(\sqrt{\frac{j\omega}{k}}x\right)$$

$$\sum_{n=1}^{\infty} C_n\int_0^l \sin\left(\frac{n\pi}{l}x\right)\sin\left(\frac{m\pi}{l}x\right)dx = -k_1\int_0^l \sinh\left(\sqrt{\frac{j\omega}{k}}x\right)\cdot\sin\left(\frac{m\pi}{l}x\right)dx - k_2\int_0^l \cosh\left(\sqrt{\frac{j\omega}{k}}x\right)\sin\left(\frac{m\pi}{l}x\right)dx.$$

$$C_m = k_1\frac{2m\pi k\cdot(-1)^m}{(m\pi)^2k + j\omega l^2}\sinh\left(\sqrt{\frac{j\omega}{k}}l\right) - k_2\frac{2m\pi k}{(m\pi)^2k + j\omega l^2}\left[1 - (-1)^m\cosh\left(\sqrt{\frac{j\omega}{k}}l\right)\right]$$

代入$k_1, k_2$ $\qquad C_m = -\frac{2m\pi kA}{(m\pi)^2k + j\omega l^2}$

$$\therefore \boxed{W(x,t) = \sum_{n=1}^{\infty}\frac{-2n\pi kA}{(n\pi)^2k + j\omega l^2}\sin\left(\frac{n\pi}{l}x\right)\cdot e^{-\left(\frac{n\pi}{l}\right)^2kt}.}$$

▲ $u(x,t) = V(x,t) + W(x,t) = \cdots$

$$W|_{t=0} = -A\frac{\sinh(1+i)\alpha(l-x)}{\sinh(1+i)\alpha l}$$

$\cos(ix) = \cosh x$

$-i\sin(ix) = \sinh$

$\frac{e^{-x}-e^{x}}{2i}$ $\quad i$

$u = X(x)T(t)$. $\quad XT' - k\cdot X''T = 0$

$$\frac{X''}{X} = \frac{T'}{kT} = -\lambda.$$

$$\begin{cases} X''+\lambda X = 0. \\ T'+\lambda kT = 0. \end{cases} \qquad \begin{cases} X(0) = 0 \\ X(l) = 0 \end{cases}$$

$\lambda = 0$ 不是特征值.

$B=0$, $A\neq 0$, $\sin\sqrt{\lambda}\,l = 0$. $\lambda = \left(\frac{n\pi}{l}\right)^2$, $n = 1, 2, 3, \cdots$

$$X_n(x) = \sin\frac{n\pi}{l}x.$$

$$T_n(t) = C_n e^{-\left(\frac{n\pi}{l}\right)^2 kt}$$

$$W|_{t=0} = \sum_{n=1}^{\infty} C_n \sin\frac{n\pi}{l}x = -A\frac{\sinh(1+i)\alpha(l-x)}{\sinh(1+i)\alpha}$$

$$= -A\frac{\sin(i-1)\alpha(l-x)}{\sin(i-1)\alpha}.$$

$$C_n = \frac{2}{l}\int_0^l -A\frac{\sin[(i-1)\alpha(l-x)]\sin\frac{n\pi}{l}x}{\sin(i-1)\alpha}\,dx.$$

$$C_n = \frac{2}{l}\int_0^l -A\,\frac{\sin[(i-1)\alpha(l-x)]\,\sin\frac{n\pi}{l}x}{\sin(i-1)\alpha}\,dx \qquad 2A$$

$$\int_0^l 2\sin(\omega'(l-x))\sin\frac{n\pi}{l}x\,dx \qquad 4$$

$$= \int_0^l \cos\left[\omega' l - \omega' x - \frac{n\pi}{l}x\right] - \cos\left[\omega' l - \omega' x + \frac{n\pi}{l}x\right] dx$$

$$= \frac{1}{+\omega' + \frac{n\pi}{l}}\sin\left[\omega' l - \omega' x - \frac{n\pi}{l}x\right]_0^l - \frac{1}{\omega' - \frac{n\pi}{l}}\sin\left[\omega' l - \omega' x + \frac{n\pi}{l}x\right]_0^l$$

$$= \frac{1}{\omega' - \frac{n\pi}{l}}\sin(\omega' l) - \frac{1}{\omega' + \frac{n\pi}{l}}\sin(\omega' l)$$

$$= \frac{2n\pi/l}{\omega'^2 - \frac{n^2\pi^2}{l^2}}\sin(\omega' l)$$

$$C_n = -\frac{1}{l}\,\frac{1}{\sin(i-1)\alpha}\,\frac{2n\pi/l}{(i-1)^2\alpha^2 - \frac{n^2\pi^2}{l^2}}\sin(i-1)\alpha.$$

$$= -\frac{2An\pi}{-2i\frac{\omega}{2k}l^2 - n^2\pi^2} = \frac{2kA\,n\pi}{(n^2\pi^2k + i\omega l^2)}$$

$$u(x,t) = A\,\frac{\sinh\left[(1+i)\sqrt{\frac{\omega}{2k}}(l-x)\right]}{\sinh\left[(1+i)\sqrt{\frac{\omega}{2k}}l\right]}e^{i\omega t} - 2kA\sum_{n=1}^{\infty}\frac{n\pi}{(n^2\pi^2k + i\omega l^2)}\sin\frac{n\pi}{l}x\,e^{-\left(\frac{n\pi}{l}\right)^2kt}$$

7. Try to solve the following solution problem

$$\frac{\partial u}{\partial t} - \kappa\frac{\partial^2 u}{\partial x^2} = 0,$$

$$u\big|_{x=0} = A\exp\left\{-\alpha^2\kappa t\right\}, \qquad u\big|_{x=l} = B\exp\left\{-\beta^2\kappa t\right\},$$

$$u\big|_{t=0} = 0.$$

*Proof.* We choose $v(x,t) = \frac{l-x}{l} Ae^{-\alpha^2\kappa t} + \frac{x}{l} Be^{-\beta^2\kappa t}$ and suppose $u(x,t) = v(x,t) + w(x,t)$. Note $v(0,t) = Ae^{-\alpha^2\kappa t}$ and $v(l,t) = Be^{-\beta^2\kappa t}$, we have the following equation for $w(x,t)$:

$$\begin{cases} \frac{\partial w}{\partial t} - \kappa\frac{\partial^2 w}{\partial x^2} = -\left[\frac{\partial v}{\partial t} - \kappa\frac{\partial^2 v}{\partial x^2}\right] = \frac{l-x}{l} A\alpha^2\kappa e^{-\alpha^2\kappa t} + \frac{x}{l} B\beta^2\kappa e^{-\beta^2\kappa t} = f(x,t) \\ w|_{x=0} = w|_{x=l} = 0 \\ w(x,0) = -v(x,0) = -\frac{l-x}{l} A - \frac{x}{l} B = \phi(x). \end{cases}$$

Solving the eigenvalue problem

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$$

gives the eigenfunctions $\{X_n(x)\}_{n=1}^\infty$, where $X_n(x) = \sin\sqrt{\lambda_n}x$ with $\lambda_n = \left(\frac{n\pi}{l}\right)^2$. Expand $f(x,t)$ according to $\{X_n(x)\}$, we have $f(x,t) = \sum_{n=1}^\infty g_n(t)X_n(x)$ with

$$g_n(t) = \frac{2}{l}\int_0^l \left[\frac{l-x}{l} A\alpha^2\kappa e^{-\alpha^2\kappa t} + \frac{x}{l} B\beta^2\kappa e^{-\beta^2\kappa t}\right] X_n(x)dx.$$

Note $\int_0^l X_n(x)dx = \frac{1}{-\sqrt{\lambda_n}} \cos\sqrt{\lambda_n}x|_{x=0}^l = \frac{(-1)^n - 1}{-\frac{n\pi}{l}} = \frac{l[(-1)^{n+1}+1]}{n\pi}$ and

$$\int_0^l xX_n(x)dx = \frac{1}{-\sqrt{\lambda_n}}\int_0^l x d\cos\sqrt{\lambda_n}x = \frac{l(-1)^n}{-\frac{n\pi}{l}} = \frac{l^2}{n\pi}(-1)^{n+1}.$$

Therefore

$$\begin{aligned} g_n(t) &= \frac{2}{l}\int_0^l \left[A\alpha^2\kappa e^{-\alpha^2\kappa t}X_n(x) + \frac{B\beta^2\kappa e^{-\beta^2\kappa t} - A\alpha^2\kappa e^{-\alpha^2\kappa t}}{l} xX_n(x)\right] dx \\ &= \frac{2\kappa}{n\pi} A\alpha^2 e^{-\alpha^2\kappa t} + \frac{2\kappa}{n\pi} B\beta^2 e^{-\beta^2\kappa t}(-1)^{n+1}. \end{aligned}$$

Expand $\phi(x)$ according to $\{X_n(x)\}_{n=1}^\infty$, we have $\phi(x) = \sum_{n=1}^\infty a_n X_n(x)$ where

$$a_n = \frac{2}{l}\int_0^l \left[-A + \frac{A-B}{l}x\right] X_n(x)dx = -\frac{2A}{n\pi} - \frac{2B}{n\pi}(-1)^{n+1}.$$

If we let $w(x,t) = \sum_{n=1}^\infty T_n(t)X_n(x)$, then

$$\begin{cases} \sum_{n=1}^\infty [T_n'(t) + \kappa\lambda_n T_n(t)]X_n(x) = \sum_{n=1}^\infty g_n(t)X_n(x) \\ \sum_{n=1}^\infty T_n(0)X_n(x) = \sum_{n=1}^\infty a_n X_n(x). \end{cases}$$

By orthogonality of $\{X_n(x)\}$, we have

$$\begin{cases} T_n'(t) + \kappa\lambda_n T_n(t) = g_n(t) \\ T_n(0) = a_n, \end{cases}$$

which implies

$$\begin{aligned} T_n(t) &= e^{-\kappa\lambda_n t}\int_0^t g_n(s)e^{\kappa\lambda_n s}ds + a_n \\ &= \frac{2\kappa}{n\pi}\left[A\alpha^2\frac{e^{-\alpha^2\kappa t} - e^{-\kappa\lambda_n t}}{-\alpha^2\kappa + \kappa\lambda_n} + B\beta^2(-1)^{n+1}\frac{e^{-\beta^2\kappa t} - e^{-\kappa\lambda_n t}}{-\beta^2\kappa + \kappa\lambda_n}\right] + a_n \end{aligned}$$

Combined, we can write the solution $u(x,t)$ to the original problem as

$$\left[\frac{l-x}{l}Ae^{-\alpha^2\kappa t} + \frac{x}{l}Be^{-\beta^2\kappa t}\right] + \sum_{n=1}^\infty \left\{\frac{2\kappa}{n\pi}\left[A\alpha^2\frac{e^{-\alpha^2\kappa t} - e^{-\kappa\left(\frac{n\pi}{l}\right)^2 t}}{-\alpha^2\kappa + \kappa\left(\frac{n\pi}{l}\right)^2} + B\beta^2(-1)^{n+1}\frac{e^{-\beta^2\kappa t} - e^{-\kappa\left(\frac{n\pi}{l}\right)^2 t}}{-\beta^2\kappa + \kappa\left(\frac{n\pi}{l}\right)^2}\right] + a_n\right\}\sin\frac{n\pi}{l}x.$$

**Remark 27.** *If we choose $v(x,t) = A\frac{\sin\alpha(l-x)}{\sin\alpha l}e^{-\alpha^2\kappa t} + B\frac{\sin\beta x}{\sin\beta l}e^{-\beta^2\kappa t}$, then it's easy to verify $\frac{\partial v}{\partial t} - \kappa\frac{\partial^2 v}{\partial x^2} = 0$. So this choice of $v(x,t)$ simultaneously homogenizes boundary condition and the differential equation, which makes the solution much easier.*

$$v(x,t)=A\frac{\sin\alpha(l-x)}{\sin\alpha l}e^{-\alpha^2\kappa t}+B\frac{\sin\beta x}{\sin\beta l}e^{-\beta^2\kappa t}。$$

令$u=v+w$，则$\begin{cases}\dfrac{\partial w}{\partial t}-\kappa\dfrac{\partial^2 w}{\partial x^2}=0\\ w\big|_{x=0}=0,w\big|_{x=l}=0\\ w\big|_{t=0}=-A\dfrac{\sin\alpha(l-x)}{\sin\alpha l}-B\dfrac{\sin\beta x}{\sin\beta l}\end{cases}$。

$$w=\sum_{n=1}^{\infty}C_n\sin\frac{n\pi}{l}xe^{-\kappa\left(\frac{n\pi}{l}\right)^2t},$$

$$C_n=-\frac{2A}{l\sin\alpha l}\int_0^l\sin\alpha(l-x)\sin\frac{n\pi}{l}xdx-\frac{2B}{l\sin\beta l}\int_0^l\sin\beta x\sin\frac{n\pi}{l}xdx$$

$$=-\frac{2A}{l}\int_0^l\cos\alpha x\sin\frac{n\pi}{l}xdx+\frac{2A}{l}\cot\alpha l\int_0^l\sin\alpha x\sin\frac{n\pi}{l}xdx-\frac{2B}{l\sin\beta l}\int_0^l\sin\beta x\sin\frac{n\pi}{l}xdx$$

$$=-\frac{2An\pi}{(n\pi)^2-(\alpha l)^2}\left[1-(-1)^n\cos\alpha l\right]-\frac{2An\pi(-1)^n}{(n\pi)^2-(\alpha l)^2}\cos\alpha l+\frac{2(-1)^n n\pi B}{(n\pi)^2-(\beta l)^2}$$

$$=-\frac{2An\pi}{(n\pi)^2-(\alpha l)^2}+\frac{2(-1)^n n\pi B}{(n\pi)^2-(\beta l)^2}。$$

上面假设不存在正整数$n,m$，使$n\pi=\alpha l$，$m\pi=\beta l$。

$$u(x,t)=A\frac{\sin\alpha(l-x)}{\sin\alpha l}e^{-\alpha^2\kappa t}+B\frac{\sin\beta x}{\sin\beta l}e^{-\beta^2\kappa t}$$

$$-\sum_{n=1}^{\infty}2n\pi\left[\frac{A}{(n\pi)^2-(\alpha l)^2}-\frac{(-1)^n B}{(n\pi)^2-(\beta l)^2}\right]\sin\frac{n\pi}{l}xe^{-\kappa\left(\frac{n\pi}{l}\right)^2t}。$$